

Aerodynamic Investigation of Cooling Drag of a Production Sedan Part 1: Test Results

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ABSTRACT

The airflow that enters the front grille of a ground vehicle for the purpose of component cooling has a significant effect on aerodynamic drag (engine airflow drag). Furthermore, engine airflow is known to be capable of influencing upstream external airflow (interference drag). The combined effect of these phenomena is commonly referred to as cooling drag, which generally contributes up to 10% of total vehicle drag. Due to this coupled nature, cooling drag is difficult to understand as it contains influences from multiple locations around the vehicle. A good understanding of the sources of cooling drag is paramount to drive vehicle design to a low cooling drag configuration. In this work, a production level Lincoln MKZ was modified so that a number of variables could be tested in both static ground and moving ground wind tunnel conditions. All tests were conducted at 80 MPH. The variables studied were: underbody shield coverage, heat exchanger resistance, cooling pack configuration, vehicle attitude, front-end sealing, exit path sealing, engine bay blockage and active grille shutter (AGS) configuration. In addition to overall vehicle drag coefficients, surface pressure taps, underbody velocity rakes and cooling pack mass flows were measured to provide better insight into the internal and external flow behaviour. This paper represents the first of a two-part series, with experimental and numerical foci respectively.

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INTRODUCTION

Automotive design challenges have grown significantly over the last few decades. Modern day vehicles must achieve the combination of: attractive styling, always-improving fuel economy, always-reducing emissions, high levels of performance and improved safety. Modern day customers also expect novel interface advancements to increase the driving experience, such as partial and one-day full-authority driver-assist technologies. Many of these design requirements inherently interfere with each other as well with areas such as packaging, weight and cost.

One of the previously mentioned attributes, fuel economy, is a measure of losses from various mechanical and chemical processes that occur in the operation of the vehicle, as well as driving habits. One of the mechanical losses is aerodynamic drag, which as [equation \(1\)](#) shows, increases exponentially with the vehicle speed and thus becomes most detrimental to fuel economy at highway speeds.

$$D = \frac{1}{2} \rho_{\infty} V_{\infty}^2 S C_d \quad (1)$$

The drag of a vehicle can only be designed for through frontal area (S) and drag coefficient (C_d). Furthermore, the drag can be divided into momentum loss around the body (external drag) and momentum loss through the body (internal drag). The dependence of these two classifications of drag on each other is referred to as interference drag

[1, 2, 3, 4]. External drag of a vehicle body is directly coupled with vehicle styling while internal drag interacts with cooling performance amongst others.

Generally, internal flow accounts for up to 10% of the total vehicle drag, which is determined through wind tunnel tests where the vehicle drag with an open and closed front grille is compared. This parameter is commonly referred to as cooling drag, shown in [equation \(2\)](#).

$$C_{d_{cooling}} = C_{d_{open\ grille}} - C_{d_{closed\ grille}} \quad (2)$$

Several strategies exist to mitigate cooling drag that have been discussed in the literature, these include: grille opening optimization, cooling fan/shroud optimization, active grille shutters (AGS), and heat exchanger ducting [5, 6, 7, 8, 9, 10, 11]. These strategies rarely only affect cooling drag, uniformity of heat exchanger flow or component thermal management may also be objectives.

The vehicle used in this study contains the AGS technology, herein AGS drag will be defined as shown in [equation \(3\)](#).

$$C_{d_{AGS}} = C_{d_{open\ vanes}} - C_{d_{closed\ vanes}} \quad (3)$$

The AGS efficiency can then be taken as the ratio between the AGS and cooling drag (4).

$$\eta_{AGS} = \frac{C_{d_{AGS}}}{C_{d_{cooling}}} \quad (4)$$

Given the complex nature of production vehicle internal air flow paths and their interactions with external air flow patterns, cooling and AGS drag have historically been difficult to understand. Several studies exist in the literature that provide insight into these phenomena. These studies can be categorized into two groups: simplified bodies and production-level bodies.

Barnard conducted experiments on a ducted Ahmed body, with adjustable flow exits in either the base or underbody [12]. His results showed that thrust recovery due to underbody flow exit angle and exit location had a significant effect on cooling drag. Williams used the Probe IV aerodynamic concept vehicle (with base region cooling flow exits) to draw relations between cooling drag and cooling system resistance [4,13]. Lai et al. used a 1/8 scale simplified notchback model to study relations between cooling drag and ground clearance, rear diffuser angle, exit-flow inclination angles and locations of the cooling flow outlet [2]. Kuthada et al. presented a simplified SAE-Body which was later modified by Baeder et al. to incorporate an underhood region and the cooling drag dependence on cooling exit angle/location, and radiator resistance were studied [14–15]. A 1/4 scale DrivAer model was modified to allow underhood flow in a collaborative effort between TU Munich and FKFS [16–17]. The DrivAer model is a generic model based on the Audi A4 and the BMW 3-series and it features interchangeable rear ends (notchback, fastback and wagonback) [18]. The open-cooling DrivAer vehicle features a simulated radiator and simplified engine block, cooling air flow is allowed to leave through either the front wheel houses and/or the underbody.

The effects of ground simulation and rear-end shape were studied by Kuthada and Wiedemann and Wickern et al. on full scale vehicles [1,19]. It was determined through an extensive testing matrix of vehicles body styles that cooling drag tends to be higher in moving ground conditions than in static ground conditions. It was theorized that because stationary wheels have a larger drag than rotating wheels, the interference flow that impacts the front wheels would have a greater impact in moving ground conditions. The resulting front tire wakes then had a downstream influence on the body style. Tesch et al. studied the influence of cooling mass flow and cooling air flow outlet location on a scale model BMW 3-series and found that for a constant cooling mass flow the cooling drag varied widely depending on the outlet location [20]. The trend of cooling drag vs. cooling mass flow was also widely different depending on the outlet location, these phenomena were attributed to different mechanisms of interference effects depending on the outlet location.

The purpose of this study is to characterize the cooling and AGS drag behaviour of a production level C-Car sedan (2013 Lincoln MKZ) with respect to design considerations that may exist during the development of a program. As it has been previously shown that

cooling drag can be influenced by the road simulation technique, select cases were studied in both stationary and moving road conditions [19].

EXPERIMENTAL SETUP

Model Details

All experiments were conducted on a modified full scale 2013 Lincoln MKZ AWD 3.7L vehicle, see Figure 1. Modifications were made to study the influence of: AGS position, AGS sealing, heat exchanger resistance, cooling pack sealing, engine bay blockage, engine bay exit paths, vehicle attitude and underbody shielding. Apart from monitoring the integral coefficients of C_d and C_p , 80 (mostly) flush surface pressure taps were mounted along the vehicle, primarily along the: upper and underbody centrelines, hood and fascia near the suspected inlet interference zones, vehicle base, heat exchangers and engine bay. This vehicle's heat exchanger configuration consists of a condenser upstream of a radiator of roughly equal size. For select tests, an array of 15 vane anemometers was placed between the heat exchangers so that aligned air flow could be measured.



Figure 1. 2013 Lincoln MKZ test vehicle, shown here in a moving ground test condition.

The engine bay blockage was varied by removing the electrical system, air-intake system, engine cover, coolant reservoir and the engine location was made adjustable with fore/aft translation, see Figure 2. Cooling pack sealing was modified in several ways, the upper sealing was controlled by removing the beauty cover (see Figure 3) and the lower sealing was controlled by the underbody engine shield.



Figure 2. Engine translation mechanism, capable of moving the engine +/- 70mm for/aft.

The AGS system was modified in two ways; (1) the AGS to fascia seals were made interchangeable so that the degree of sealing could be adjusted, and (2) the AGS vanes were modified so that each quadrant could be moved forward or rearward independently, see Figures 4 and 5 for illustrations.

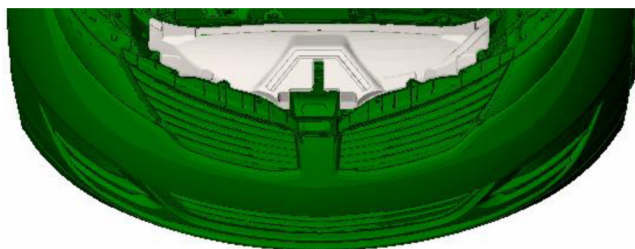


Figure 3. Beauty cover geometry, shown in white.

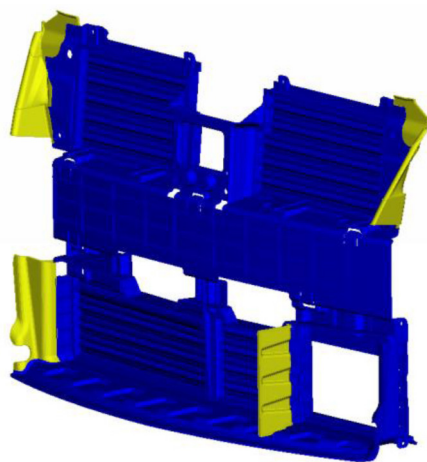


Figure 4. Side seals (yellow) that were interchangeable so that various percentages of sealing could be simulated. Above image is used for concept illustration purposes only and is not the physical article that was tested.

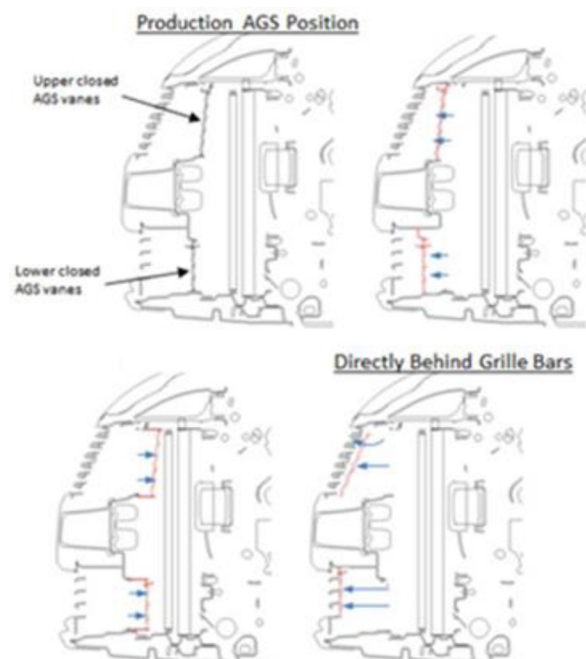


Figure 5. Adjustable AGS position, possible vane locations extended to directly in front of the condenser to directly behind the grille bars.

As previously mentioned, this vehicle features a condenser upstream of a radiator of roughly equal size. Each the condenser and radiator were made removable and perforated plates of various porosity could be added to the reverse side of the radiator. This setup allowed for the influence of heat exchanger presence and resistance on engine bay air flow to be tested. The available condenser and radiator configurations are shown in Figures 6 and 7 respectively.



Figure 6. Condenser configurations, (1) without core, (2) with core.

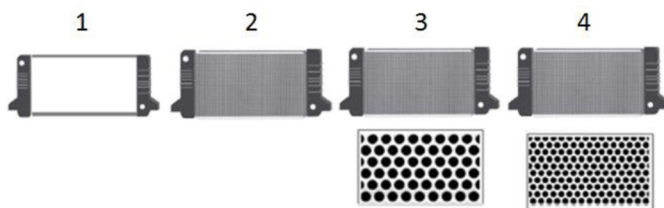


Figure 7. Radiator configurations, (1) without core, (2) with core, (3) core and additional porous plate, (4) core with additional porous plate with higher porosity.

An underbody shield package was constructed so that a completely flat, sealed underbody could be tested, and so that the air flow exit paths could be controlled through the underbody. Figure 8 shows the shield package that consists of: a rear diffuser shield, rear suspension shields, a centre tunnel shield, rocker shields and an engine shield. To force the engine bay air flow through the underbody of the vehicle the wheel liners could be sealed. Due to the complex geometry of the engine bay it was not pursued to perfectly seal the tunnel region in order to force the engine bay flow through only the wheel arches.

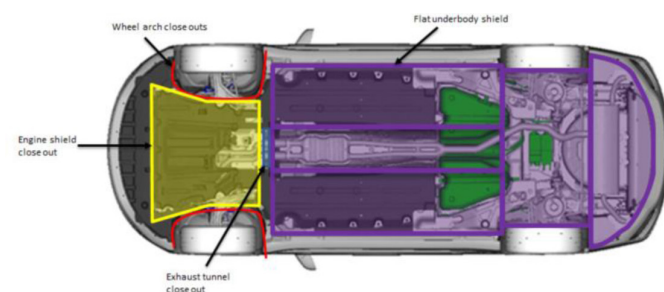


Figure 8. Flat underbody shield package where each shield could be removed individually.

The vehicles suspension was replaced with adjustable screw jacks so that vehicle attitude could be set consistently and so that attitude variation as a result of wind load could be minimized.

Test Facility Details

All static ground tests were carried out at the Ford DTF Wind Tunnel 8 (WT8) facility. WT8 is a full scale, $\frac{3}{4}$ open jet, closed return design with a 16.7 m² nozzle and a top speed of 54 m/s (120 mph). The Boundary Layer Control System (BLCS) consists of a tangential

blowing slot positioned just inside of the nozzle exit plane. The free stream turbulence intensity is below 0.4%. All presented data from this facility does not include correction methods [21].

All moving ground tests were carried out at Windshear Inc. (WSI). WSI is also a full scale $\frac{3}{4}$ open jet, closed return design with a 16.7 m² and a top speed of 80 m/s (180 mph). Ground simulation is achieved with a wide belt moving ground plane. The BLCS consists of a primary suction scoop and two tangential suction slots, positioned immediately upstream of the moving belt leading edge, inside of the nozzle exit plane. Free stream turbulence intensity is below 0.3%. All data presented from this facility has the Mercker-Wiedemann corrections (blockage and pressure gradient/buoyancy) applied [22].

RESULTS

All experiments were carried out under assumed adiabatic flow conditions with a Reynolds number of 1.06e7 on a stationary ground unless otherwise mentioned. All AGS experiments refer to comparisons between fully open and fully closed AGS positions, intermediate AGS increments were not studied. Herein the baseline configuration refers to cases with the production level AGS system and shields and the attitude set to the curb ride heights.

AGS Side Seals

Figure 9 shows the effect of varying the size of the AGS side seals on AGS efficiency. An AGS side seal gap percentage of zero percent refers to the condition where no side seals are attached to the AGS frame. For all experiments, the top, bottom and central portions of the AGS were sealed. Also for the closed AGS experiments the AGS vanes were replaced with a single surface which represented all of the vanes for each quadrant, this ensured that no AGS vane gaps occurred due to wind load. An AGS side seal gap percentage of 100 refers to the case where seals were used that had no gap between the AGS frame and either fascia or grille and complete AGS system sealing was assumed. Three sets of seals of different dimensions were interchanged to produce Figure 9. The seal in each AGS quadrant was different due to packaging complexity, the largest seals though, were on the order of 2.5" wide.

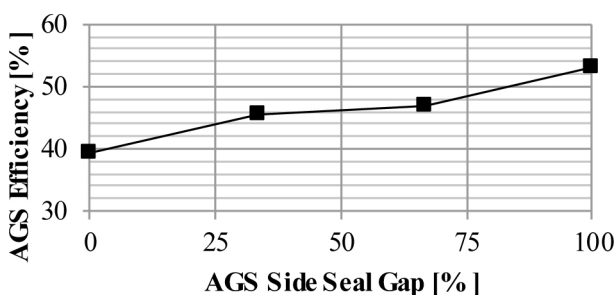


Figure 9. Trend between AGS efficiency and side seal gap percentage.

The trend in Figure 9 nearly linearly shows that as sealed area is increased (or leakage area is decreased) AGS efficiency is improved. An AGS efficiency increase of roughly 15% was observed for the

conditions tested. Another interesting conclusion is that even with perfect sealing the AGS efficiency only achieved a value of roughly 55%.

AGS Vane Position

Figure 10 shows the effect of AGS vane fore/aft position on AGS efficiency. Vane fore/aft position is measured from the front face of the condenser. In addition, the upper and lower vanes were moved independently of each other. Each series in Figure 10 represents a single position for the upper vanes where the lower vanes positions were varied from the front face of the condenser (0mm) to the back side of the grille (170mm).

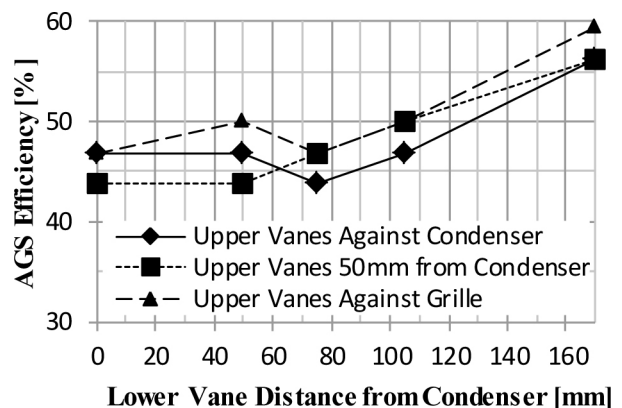


Figure 10. AGS efficiency as a function of vane position relative to the condenser.

There is no appreciable trend for any of the series between lower AGS vane locations of 0-105mm in front of the condenser. Between 105 and 170mm however, all of the curves show a clear positive trend suggesting that regardless of upper vane position, lower vane position can provide an AGS efficiency benefit if they are positioned within 65mm of the grille.

It is generally expected that the relations shown in Figure 10 would not be valid across car lines, or potentially across different AGS system designs. For this particular vehicle it is known that there is a flow bias towards the lower grille opening and thus it was expected that the lower AGS vanes positions would have a greater influence than that of the upper vanes, whose fore/aft position showed almost no influence on AGS efficiency.

For further clarification, the measured Cp values on the vehicle underbody centerline are shown in Figure 11. In this figure, the upper AGS vanes were left in the production position and the lower vanes were moved from the condenser to the grille. It can be seen that as the lower vanes are moved forward, the magnitude of negative pressure increases near the front lip of the vehicle. This suggests that as the lower AGS cavity is reduced in size, the flow that travels under the vehicle is directly affected. Among the remaining surface pressure taps situated around the vehicle, no other locations showed significant dependence on AGS vane position. Thus, no surface other than the front lip of the lower fascia was adjudged to be experiencing significant interference drag.

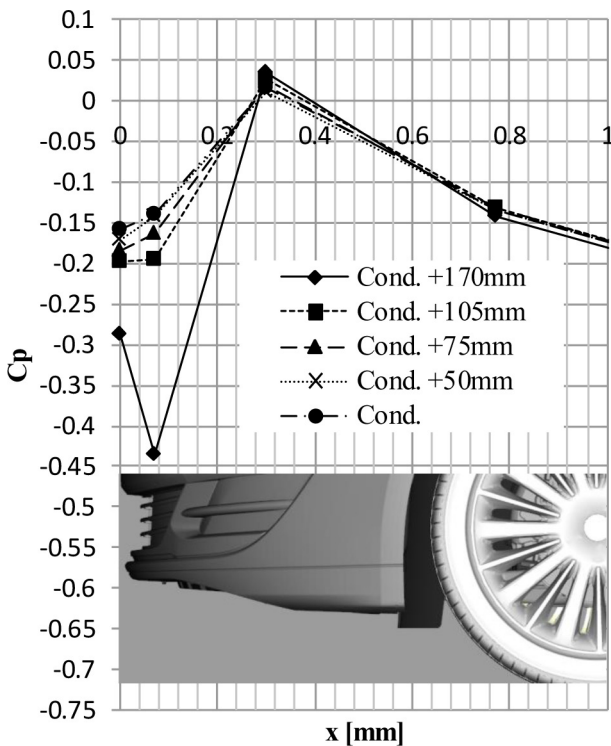


Figure 11. C_p values for the underbody centreline with the lower AGS vanes being moved from the condenser to the grille. Upper AGS vanes were left in the production position.

Vehicle Attitude

As mentioned earlier the vehicles suspension was replaced with screw jacks for (1) easy and consistent ride height adjustments, and (2) to minimize the vehicle attitude variation due to the aerodynamic force on the vehicle due to the external wind load. Vehicles with significant front or rear lift values can experience ride height shifts depending on wind tunnel velocity, furthermore, vehicle roll can be inadvertently induced during yawed conditions.

Figure 12 shows the cooling and AGS C_d deltas as functions of changes in ride height from the curb position on the left axis and the open AGS volume flow rates, also as deltas from the curb ride height on the right axis. It can be clearly seen that vehicle attitude has an influence on the closed grille condition and the effectiveness of the AGS. In normal modes of spatial payload distribution of front-engined vehicles however, the two rightmost data points in Figure 12 are considered rare as they represent downwards pitched attitudes. When considering unwanted pitch due to wind loads, typical sedan (without specific lift generating devices) front ride height raises are general well less than 10mm. Thus, it can be interpolated from the first point in Figure 12 that variations in cooling and AGS behaviour due to wind tunnel pitch would be on the order of 0.000-0.004 C_d . The volumetric flow rate curve shows that although the cooling drag changes up to 0.007 C_d , the volumetric flow rate change through the condenser is near zero. Thus, for this vehicle, the cooling drag variations as a result of ride height modifications are attributed to interference flow and not cooling pack/engine bay air flow. Though

not shown here, an analysis of local C_p data was done (similar to that of Figure 11) and again, aside from the front lip, no significant areas of interference drag were identified.

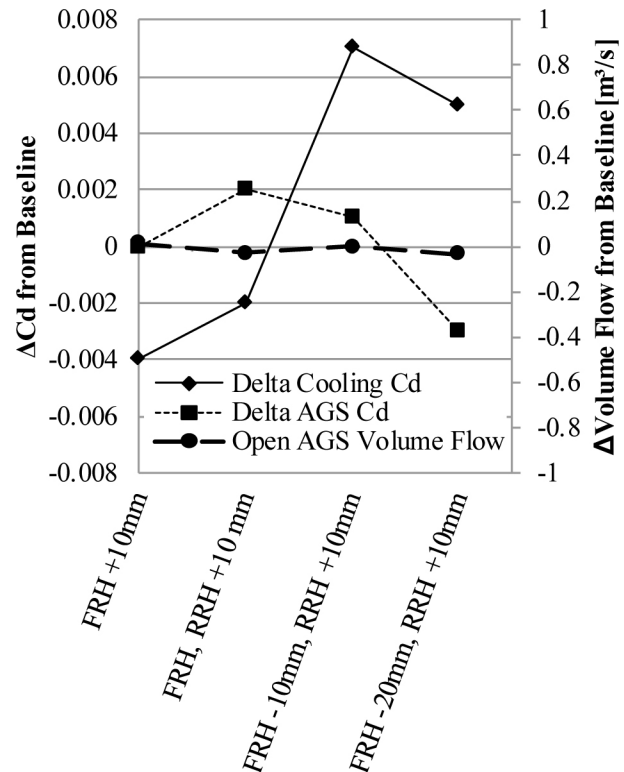


Figure 12. Delta Cooling and AGS C_d and AGS open volume flow values for varying ride heights, shown as deltas from baseline (increases in drag/volume flow are positive). FRH (Front Ride Height), RRH (Rear Ride Height).

Engine Bay Blockage and Porous Media

Three separate studies were conducted to help understand the influence of engine bay blockage:

1. Blockage influence of radiator and downstream components with condenser present.
2. Blockage influence of radiator and downstream components without condenser present.
3. Blockage influence of downstream components without radiator or condenser present.

Here, downstream components refers to: AIS (air intake system), electrical sub-system (battery, fuse box), coolant overflow tank and engine cover. Each of these studies were repeated with the engine underbody shield removed as it was known that this shield was highly influential to cooling drag.

Figure 13 shows that with the engine shield (ES) on, increasing the radiator flow resistance increased the cooling and AGS drag by a factor of 0.000-0.002 C_d . With the engine shield removed the trend is the opposite, the added radiator plates yielded lower cooling and AGS drags by the same 0.000-0.002 C_d . The un-shielded underbody is the most direct and largest leak path for engine bay air flow and thus, it stands to reason that when radiator resistance is increased and no engine shield is present, air flow is free to bypass the radiator and

leave through the underbody. With the radiator core completely removed, the change in cooling and AGS drag is also only 0.000-0.002 Cd. When further components were removed from the engine bay, a change in cooling and AGS drag was again, only seen if the engine shield was present, otherwise flow was allowed to exit freely through the underbody. It is perhaps most unexpected that the removal of the radiator core only resulted in a change of 0.000-0.002 Cd. It is theorized here, that because the radiator is the most downstream of a two heat exchanger configuration that the momentum loss was almost entirely experienced in the first exchanger. This hypothesis was further tested in studies (2) and (3).

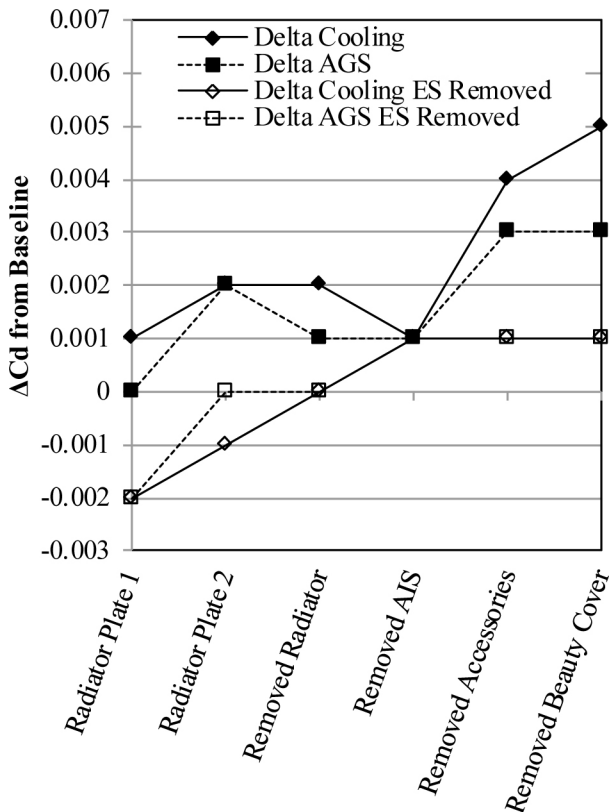


Figure 13. Delta cooling and AGS Cd for engine bay blockage study (1 – Blockage influence of radiator and downstream components with condenser present), shown as deltas from the baseline configuration (increases in drag are positive). Note that the accessories were removed while no radiator was present.

Figure 14 shows engine bay blockage study (2), where the condenser was removed and the radiator remained present. The direct influence of the removal of the condenser was only 0.001-0.002 Cd for the AGS drag delta, though it ranged from 0.001-0.005 Cd for the cooling drag delta. A larger influence than what was seen with study (1) was expected because here the most upstream heat exchanger was removed, with the condenser removed more leak paths were opened before the flow impacts a heat exchanger. As with study (1), equal or larger influences of downstream components on cooling and AGS drags were seen with the engine shield on, as opposed to off (with the AIS case being the lone exception). It may have been expected that with the condenser removed, direct changes to the radiator resistance (added radiator resistance plates) would show a larger impact on either cooling or AGS drag. Figure 14 however, shows that the

radiator plates only altered those metrics again by 0.000-0.002 Cd. This result may be due to the fact that additional side leak paths were opened up when the condenser was removed and air flow that would have otherwise experienced greater resistance with the added radiator plates would have travelled through these leak paths instead. The effect of removing downstream accessories was similar to that of study (1). The trend regarding the presence of the engine shield remains, being that each engine bay components influence was reduced as the engine shield was removed.

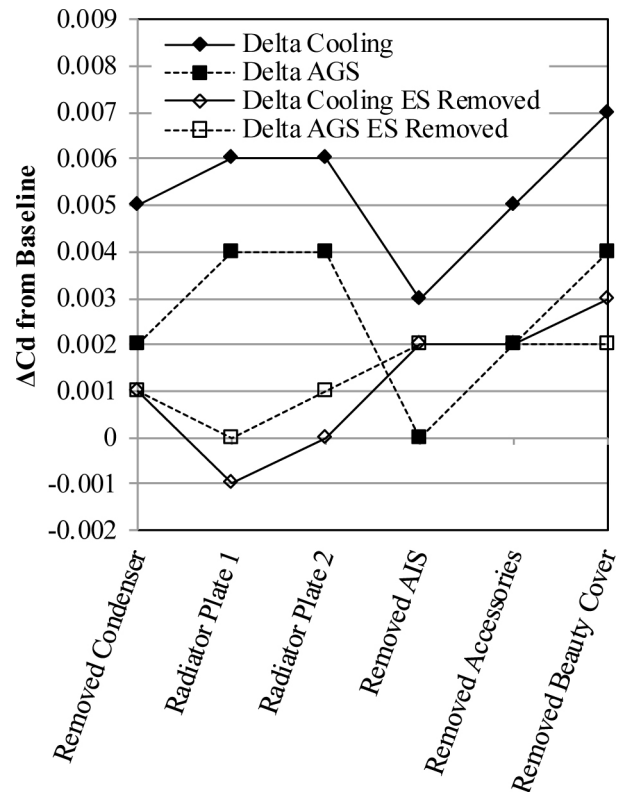


Figure 14. Delta cooling and AGS Cd for engine bay blockage study (2 – Blockage influence of radiator and downstream components without condenser present), shown as deltas from the baseline configuration (increases in drag are positive). Note that for last three points the radiator plates were removed.

The results of engine bay blockage study (3) are shown in Figure 15. A clear influence of the heat exchanger presence is seen in the first three points, however the engine shield influence is much more prevalent for the cooling drag delta, from the previous conclusions this was expected. The influence of downstream accessories was previously shown to influence cooling and AGS drag, Figure 15 shows however that in the case where no heat exchangers are present, the presence of the tested downstream accessories had almost no significance. It is assumed that because this vehicle has a dominant amount of cooling air flow travelling through the lower grille opening that with no heat exchangers present the air flows preferred exit paths would also be biased towards lower leak paths (below engine, tunnel, wheel arches). Each accessory that was removed was situated in the upper half of the engine bay and possibly for this reason, did not have influence on the velocity cooling air flow which was made much higher as a result of the heat exchangers being removed.

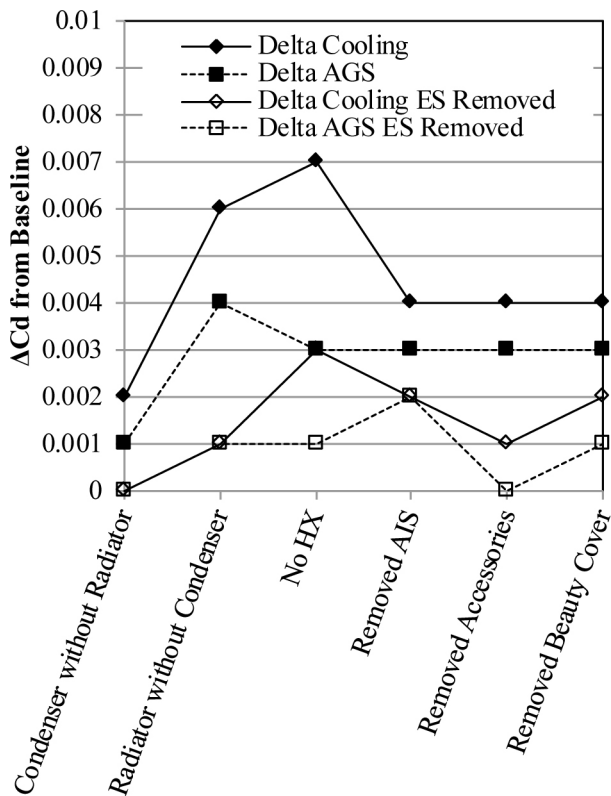


Figure 15. Delta cooling and AGS Cd for engine bay blockage study (3 – Blockage influence of downstream components without radiator or condenser present), shown as deltas from the baseline configuration (increases in drag are positive).

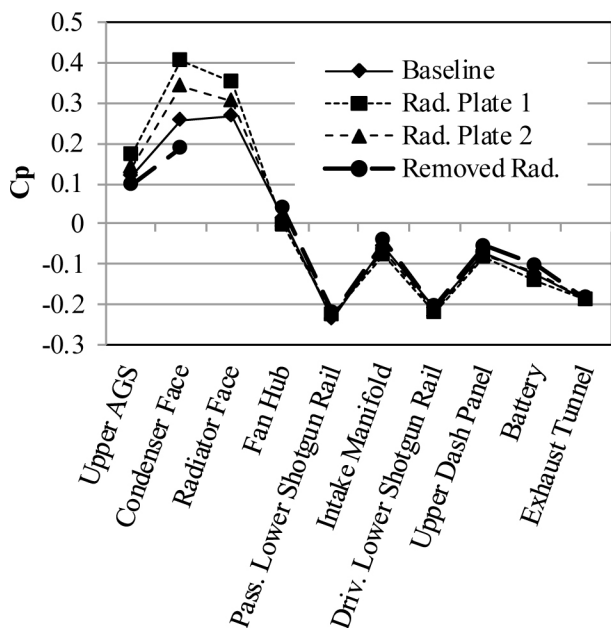


Figure 16. Local Cp measurements in various locations around the engine bay with open AGS conditions and varying radiator conditions.

Further investigations were made on select configurations of study (1) with local Cp and anemometer measurements (Only study (1) was chosen because the anemometer array was fixed to the reverse side of the condenser). Figure 16 shows that the pressures upstream of the radiator are influenced by the increase or decrease of the radiator resistance but none of the tested locations downstream see any appreciable difference. Also, the pressures upstream of the radiator are affected but the region affected does not influence the exterior flow, as shown by exterior Cp measurements (not shown here). Figure 17 shows the volumetric air flow measurements and cooling and AGS drags as a function of radiator resistance configurations and it can be seen that large changes in volumetric air flow are required to influence cooling and/or AGS drag.

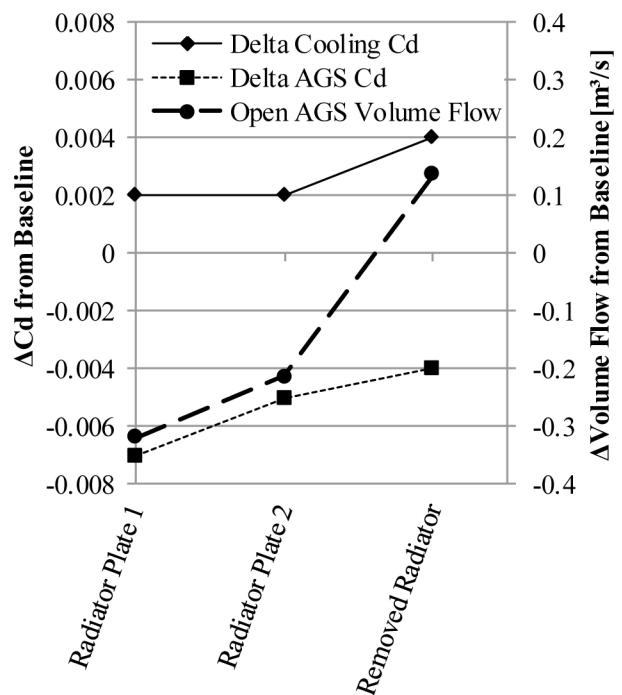


Figure 17. Delta cooling and AGS drags and volume flow rates shown as deltas from the vehicle baseline (increases in drag/volume flow are positive).

Underbody Shields

The presence of underbody shields and their influences were studied in both stationary and moving ground conditions. Each shield was removed sequentially from the rear to the front of the vehicle. The shields used in this particular study formed a flat underbody (aside from closing the diffuser and closing ducts in the engine shield), for visual reference see Figure 8. The flat shields consisted of: a diffuser shield, rear suspension shields, a central tunnel shield, rocker shields and the production engine shield with covered cooling ducts. Also included in the study were modified front wheel arch liners/shields which blocked air flow from exiting through the wheel arches. Additionally, a vertical tunnel cover at the dash panel which blocked flow from travelling through the upper half of the tunnel was used.

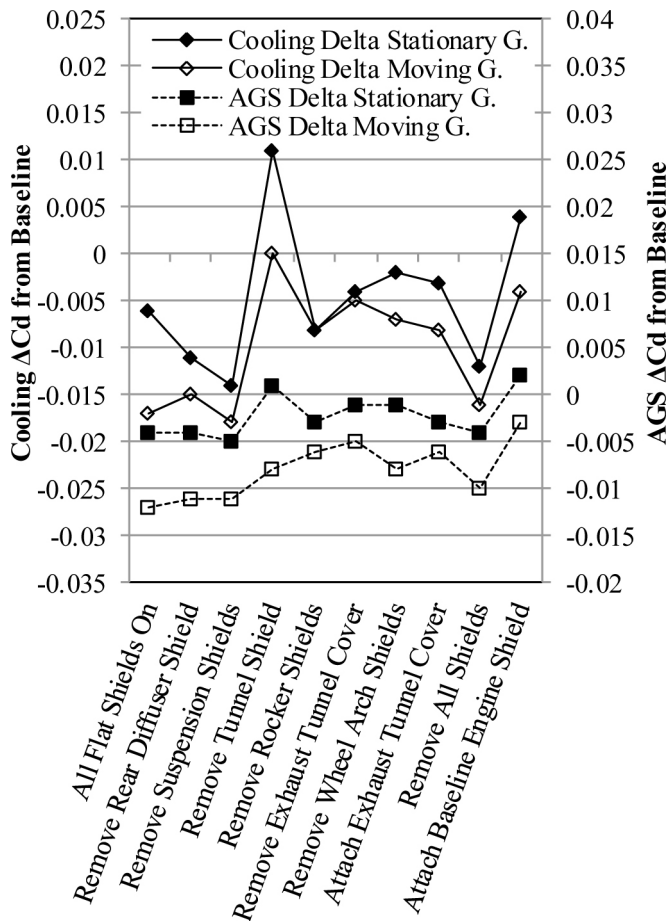


Figure 18. Delta cooling and AGS Cd for flat underbody shield study, shown as deltas from the baseline (increases in drag are positive).

Figure 18 shows the results for the complete study done for both cooling and AGS drag. For all configurations shown, the cooling and AGS drag deltas ranged from 0.000-0.011 and 0.003-0.008 Cd lower on moving ground conditions respectively. The AGS delta trends in general match between stationary and moving ground conditions, however, divergence exists when the rocker shields and exhaust tunnel shields are removed from their respective configurations. Each shields influence on cooling drag yielded the same trend in static or moving ground conditions aside from the rear diffuser shield and the wheel arch shields whose trends were divergent. The sharp rise in cooling drag seen when the centre tunnel shield is removed can be attributed to the decrease in base pressure as measured by base surface pressure taps. In this configuration, all of the cooling air flow exits through the tunnel and the underbody flow and vehicle wake are affected significantly. Figure 19 shows the result of equation (5) (case x = Removed Suspension Shields, case y = Removed Tunnel Shield) evaluated at each pressure tap location (shown in Figure 20) over the base of the vehicle. When the rocker shields are then removed the base pressures return and cooling drag return to relatively lower values. The exhaust tunnel cover was tested with and without sealed front wheel arches and its presence was found to consistently reduce cooling drag.

$$\Delta C_{p_{cooling_{case\ x}}} - \Delta C_{p_{cooling_{case\ y}}}$$

(5)

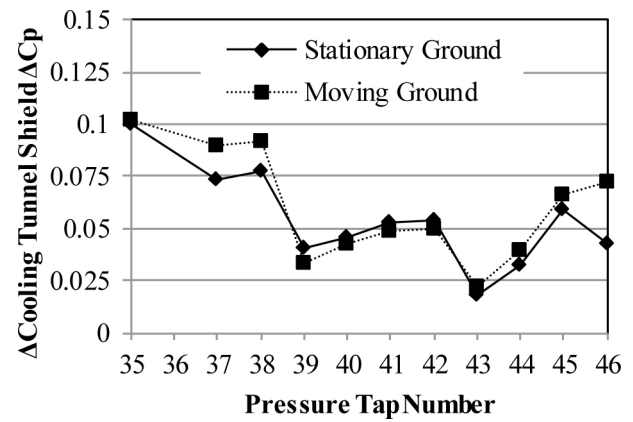


Figure 19. Δ Cooling Tunnel Shield ΔC_p for base surface pressure taps in both stationary and moving ground conditions, using equation (5) case x = Removed Suspension Shields, case y = Removed Tunnel Shield.



Figure 20. Surface pressure tap locations on base of vehicle.

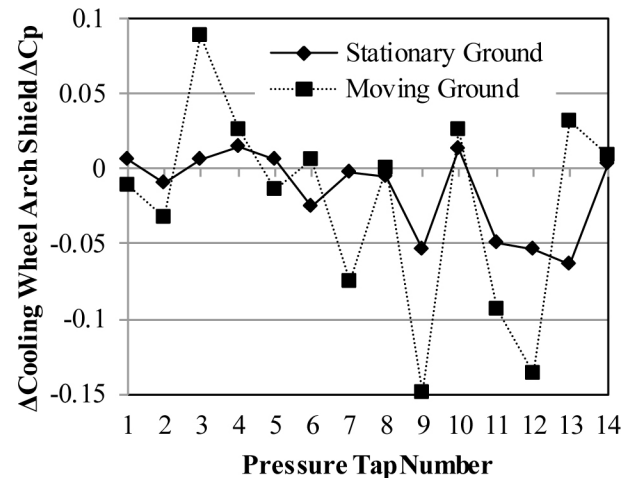


Figure 21. Δ Cooling Wheel Arch Shield ΔC_p for wheel liner/vicinity surface pressure taps in both stationary and moving ground conditions, using equation (5) case x = Removed Exhaust Tunnel Cover, case y = Removed Wheel Arch Shields.

As mentioned before, the presence of wheel liner shields reduced cooling drag with moving ground conditions and increased cooling drag with stationary ground conditions, this can possibly be attributed to the largely different C_p trends measured in each test condition in/ around the wheel liners. Equation (5) is again used (case x = Removed Exhaust Tunnel Cover, case y = Removed Wheel Arch Shields). Figure 21 shows that no appreciable trend exists between the stationary and moving ground ΔC_p values, Figures 22 and 23 show the surface pressure tap locations referenced.

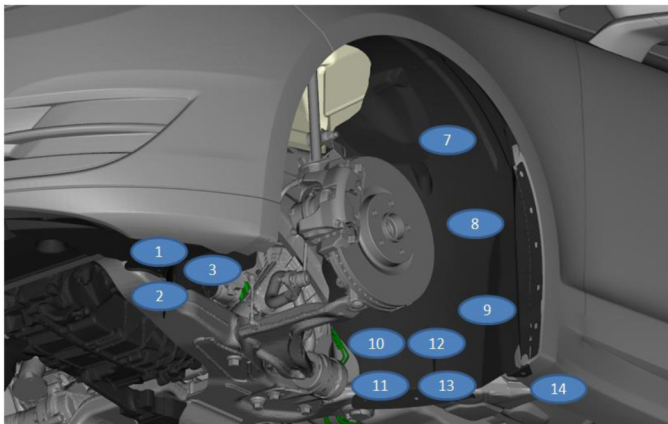


Figure 22. Wheel liner/vicinity surface pressure tap locations used for Figure 21.

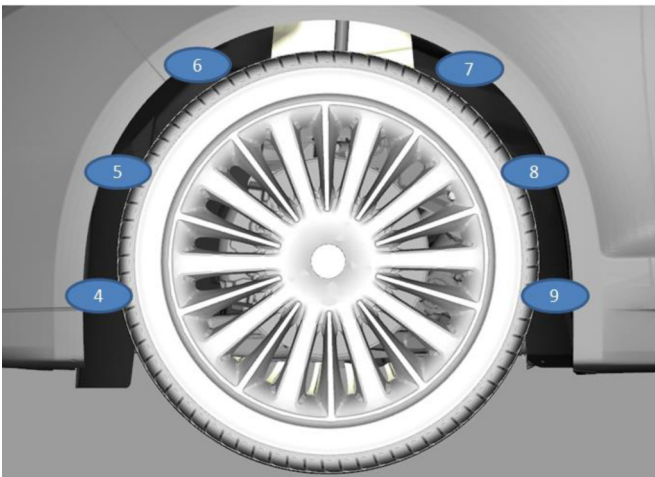


Figure 23. Wheel liner surface pressure tap locations used for Figure 21.

CONCLUSION

Several studies were presented within regarding the variation of both cooling and AGS drag deltas with respect to various component changes. Not only was it shown that non-cooling related components such as underbody shields could have a significant impact on these metrics, but it was shown that the impact of non-cooling related components could be larger than that of direct cooling flow related components (AGS seals etc). It was shown that cooling and AGS drag tend to be lower in moving ground conditions, however, the

extent of the difference can vary depending on the vehicle configuration. The cooling and AGS drag dependencies on cooling pack configuration and engine bay packaging was studied and it was concluded that the most upstream heat exchanger (given relative size considerations) has the largest influence.

Although this work sheds more light on the nature of cooling air flows and their interaction with external air flows, the topics of cooling and AGS drag each remain considerably complex and further investigation is required to determine if the conclusions made within apply to either; other sedans or other vehicle shapes.

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